

Two winning piles can make a loss

Name: _____

Date: _____

Use moves 1, 3, 4. Now there are **two piles**; a turn changes exactly one pile. The player who takes the very last counter wins.

1. W and L forget too much.

Play from piles (1,1), then (1,4). Each individual pile is W. Are both combined games W? Explain your best play.

2. Give each pile a richer label.

Let $g(0) = 0$. For a larger pile, list the g -values reachable in one move; give the pile the **smallest nonnegative integer missing** from that list. For example, $g(1) = 1$ and $g(2) = 0$.

Pile	0	1	2	3	4	5	6	7	8	9	10
g	0	1	0								

3. Test an equality rule.

Conjecture: two piles are losing together exactly when their g -values are equal. Check (1,3), (4,6), and (5,6). Find a winning move wherever possible.

Checks and moves:

4. Prove the rule for two piles.

Why can a move never keep equal labels equal? If labels differ, why can you move from the larger label to the smaller one? Use the “smallest missing” definition.

The two parts of my proof:

Beyond this page: Three piles require binary XOR, not ordinary addition or pairwise equality. These labels are Sprague–Grundy values.

Build first. Explain by pointing, talking, or drawing.